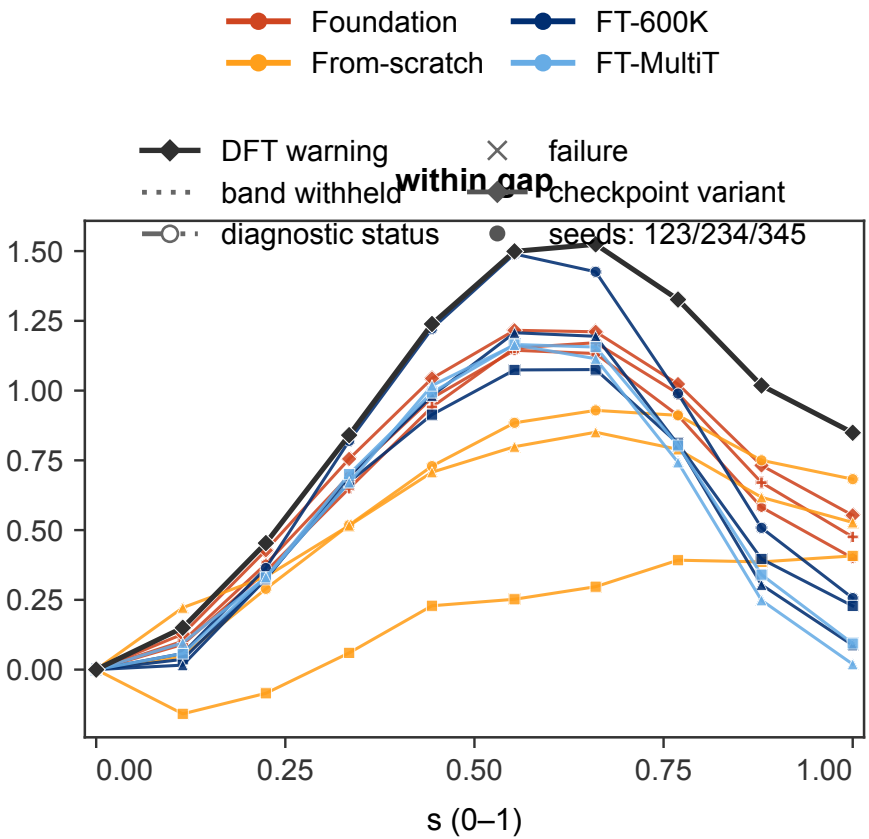ed spheres = real retained path-image overlay | α = image order only, not un